SODIUM & WATER METABOLISM & HOMEOSTASIS

Dr. Muhammad Ahsan

Normal Metabolism of Water and Sodium

Some basic concepts:

1. Homeostasis

- Homeostasis indicates a stable internal environment within the body, which means:

- (a) body fluid: normal volume, composition and pH ;
- (b) normal temperature, blood pressure, cardiac output;
- (c) normal level of hormones, et al.

Normal body fluid is the most important part in the homeostasis.

2. Body fluid (

Body fluid is the fluids in the body.

Body fluids are solution with

solutes (electrolytes, glucose, amino acids, fatty substances and other constituents) and

solvent (water).

(1)Volume and distribution of body fluid

Intracellular fluid

- Body fluid (40%)ICF
- (60% of body IVF(5%)
  weight)

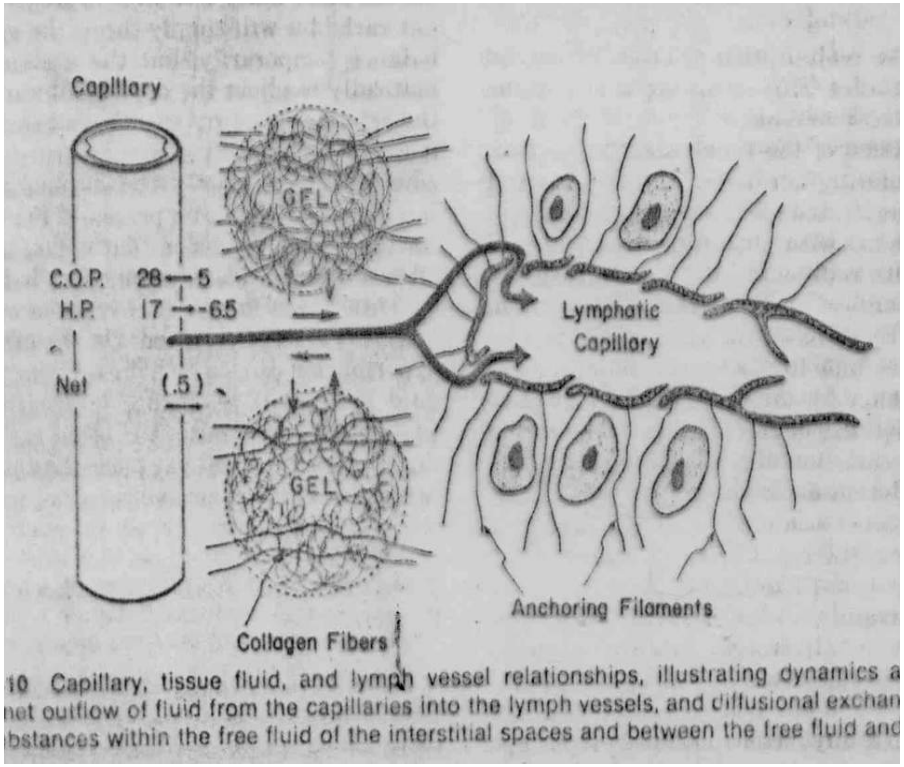
- Extracellular fluid
- (20%)ECF EVF (15%)
- Body fluid constitutes 60% of body weight.
-

ECF includes:

IVF: plasma, 5%

EVF: 15%

(1)interstitial fluid,
(2)lymph fluid
(3)transcellular fluid.



10 Capillary, tissue fluid, and lymph vessel relationships, illustrating dynamics a
net outflow of fluid from the capillaries into the lymph vessels, and diffusional exchan
bstances within the free fluid of the interstitial spaces and between the free fluid and

The transcellular fluid is the fluids in cavities:

cerebrospinal fluid (in ventricles of brain),

pleural fluid (in pleural cavity),

pericardial fluid (in pericardial cavity)

joint fluid (in articular cavity)

Transcellular fluid comes from epithelial cells.

99% of interstitial fluid is fixed to collagen, mucopolysaccharide and hyaluronic acid (gel), (connective tissue), which called fixed water.

1% of interstitial fluid is free water (moving freely).

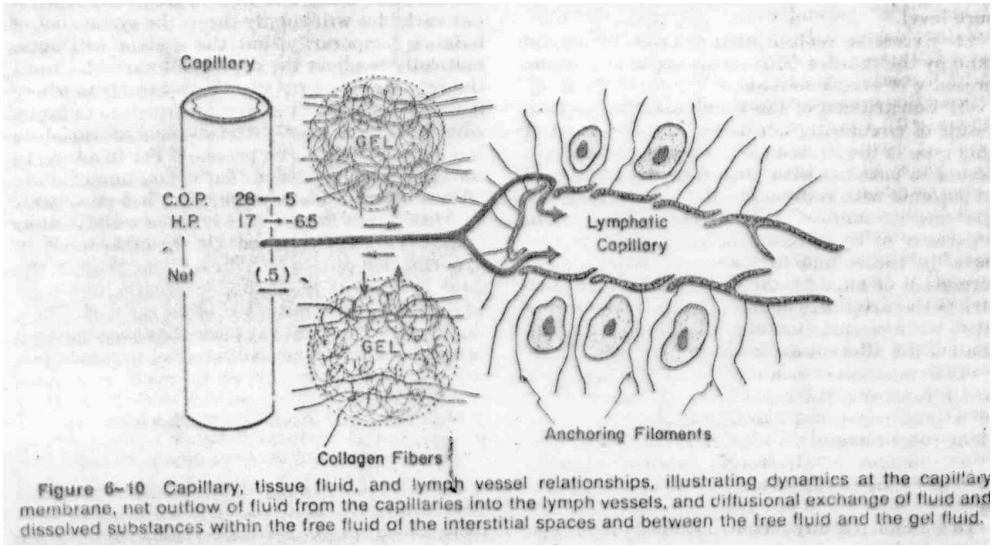


Figure 6-10 Capillary, tissue fluid, and lymph vessel relationships, illustrating dynamics at the capillary membrane, net outflow of fluid from the capillaries into the lymph vessels, and diffusional exchange of fluid and dissolved substances within the free fluid of the interstitial spaces and between the free fluid and the gel fluid.

Influential factors on the volume of body fluid

age	% of BW
new born(0~1mon)	80%
infant (1~12mon)	70%
school age(1~17 year)	65%
adult	60%

Influential factors on the volume of body fluid

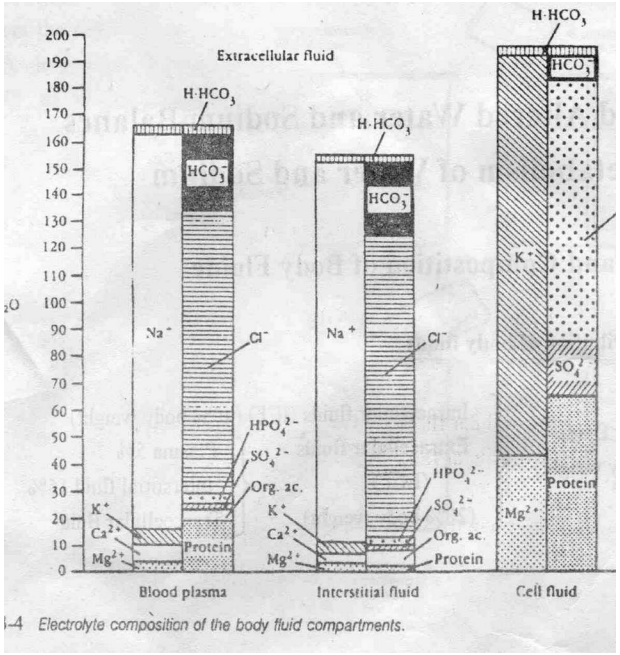
organ (tissue)	water content
fat	25%~30%
muscle	76%
bone	14%~46%
liver	70%
skin	72%

A fatty person and a thin person with the same body weight lost the same volume of body fluid, whose condition is worse?

(2)Composition of body fluid

1) plasma (ECF)

Cations (mEq/L)		Anions (mEq/L)	
Na^+	142	$HCO3^-$	27
K^+	5	Cl^-	103
Ca^{2+}	5	$HPO4^{2-}$	2
Mg^{2+}	2	$SO4^{2-}$	1
		organic acid	5
		proteinate	16
total	154		154



Electrolyte composition of the body fluid compartments.

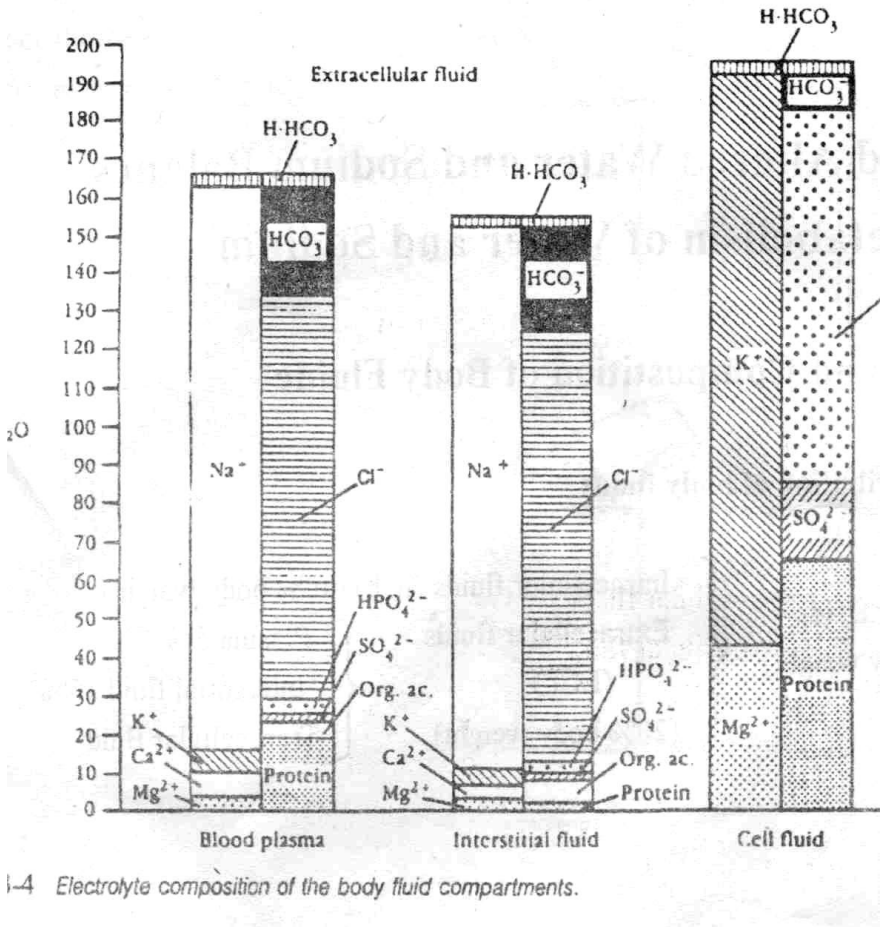
(a)The major cation is Na^+ , which cannot be replaced.

(b) The major anions are Cl^- and $HCO3^-$.(can be replaced from each other.

(c) Electrical neutral is present according to the numerals of mEq/L in the normal state.

2)interstitial fluid

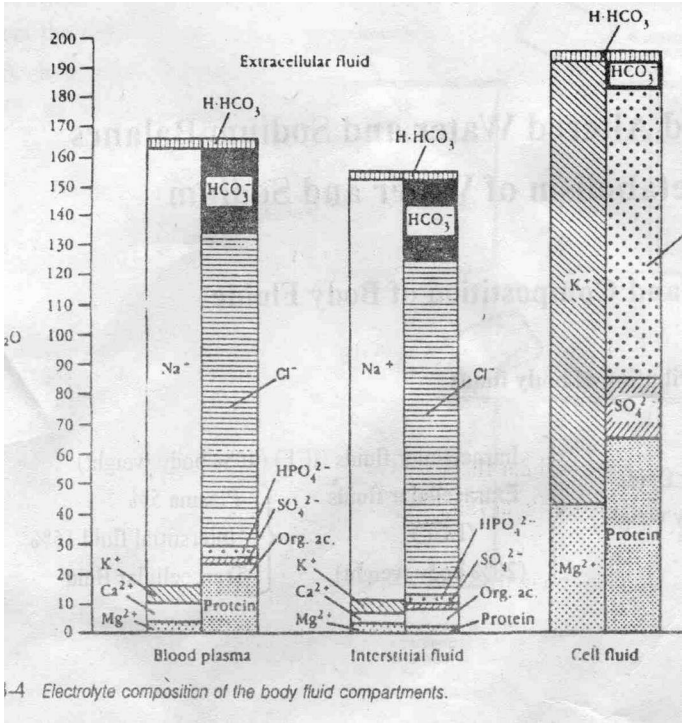
The main difference of composition between plasma and interstitial fluid is the protein concentration in interstitial fluid.



1-4 *Electrolyte composition of the body fluid compartments.*

3)ICF

Cations(mEq/L)		Anions(mEq/L)	
Na^+	15	$HCO3^-$	10
K^+	150	CI^-	1
Ca^{2+}	2	$HPO4^{2-}$	100
Mg^{2+}	27	$SO4^{2-}$	20
		proteinate	63
total	194		194



Electrolyte composition of the body fluid compartments.

Characteristics:

(1)The major cation is K^+ , which cannot be replaced.

(2) The major anions are phosphate($HPO4^{2-}$)and sulphate ($SO4^{2-}$).

(3) Electrical neutral is present according to the numerals of mEq/L in the normal state.

(3) Osmotic pressure

- Osmosis is the net diffusion of water through a selectively permeable membrane

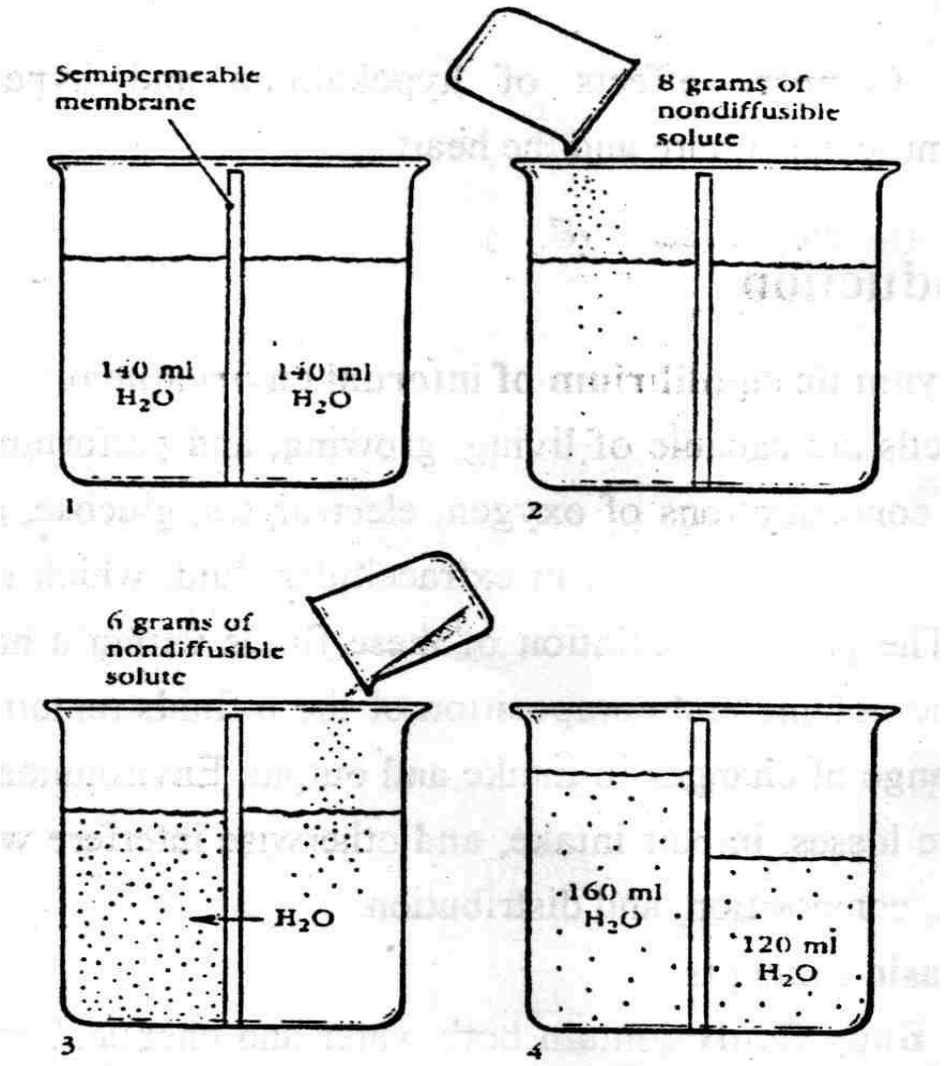


Figure 2-1 *Osmosis.* ***A.*** *A beaker of water with superimpermeable membrane separating the sides.* ***B.*** *Add 8 g of a nondiffusable substance to one side.* ***C.*** *Add 6 g of a nondiffusible substance to the other side.* ***D.*** *The water moves toward the more concentrated side and makes the concentrations equal, with more fluid on one side than the other.*

- The osmotic pressure is formed on the membrane as the water moves from the sides with lesser number of non-diffusible particles to the side with greater number of non-diffusible particles.
-
- The osmotic pressure pulls the water to the side with more number of non-diffusible particles.

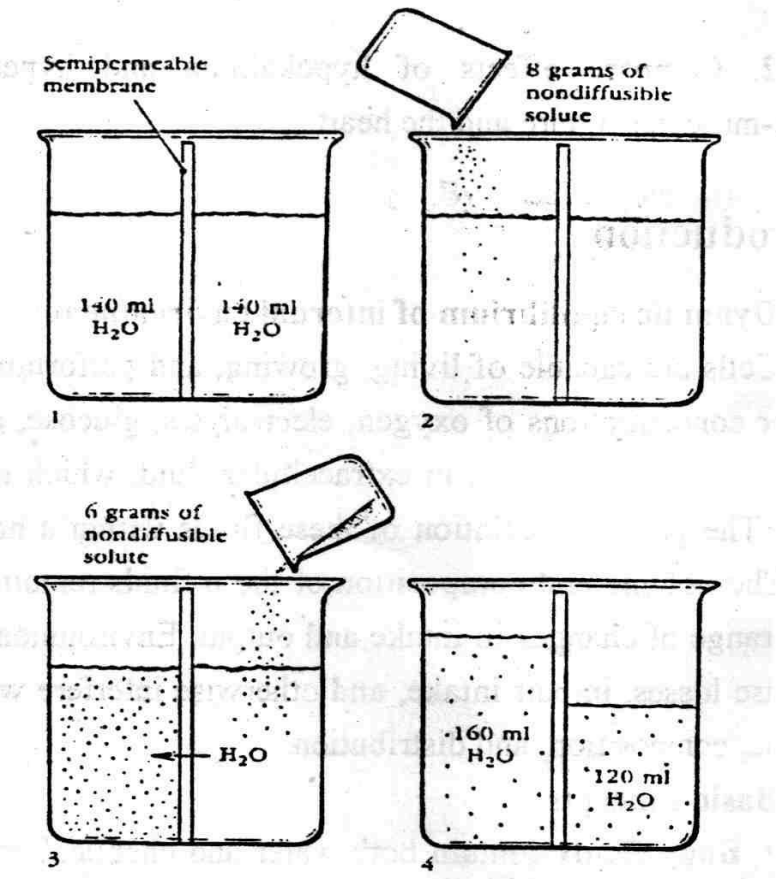


Figure 2-1 *Osmosis.* ***A.*** *A beaker of water with superimpermeable membrane separating the sides.* ***B.*** *Add 8 g of a nondiffusable substance to one side.* ***C.*** *Add 6 g of a nondiffusible substance to the other side.* ***D.*** *The water moves toward the more concentrated side and makes the concentrations equal, with more fluid on one side than the other.*

- The osmotic pressure is decided by the number of particles. It is not determined by the molecular weight or electric charge of the particle..
- The greater the number of non-diffusible particles in the solution, the greater its osmotic pressure is.
- Question: Do a Na^+ and a molecule of protein form the same osmotic pressure?

- The total osmolality is 280 ～ 320 mOsm/L (average 300 mOsm/L) in body fluid.
- 90% ～ 95% of osmotic pressure is determined by electrolytes (anions+cations).
- The anions (negative ions) constitute 139 mOsm/L.
- The cations (positive ions) constitute 151 mOsm/L.
- Glucose, urea, etc constitute 10 mOsm/L.

Classification of osmotic pressure

- The crystal osmotic pressure is mainly formed mainly by particles of electrolytes.
-
- The colloidal osmotic pressure is formed by particles of proteins.

(4) Characters of isotonicity
H_2O exchange across cell membrane

The osmotic pressure in the cells and outside the cells usually are the same.

There is a tendency to keep the isotonicity between intracellular and extracellular spaces.

Which type of molecule move for keeping isotonicity?

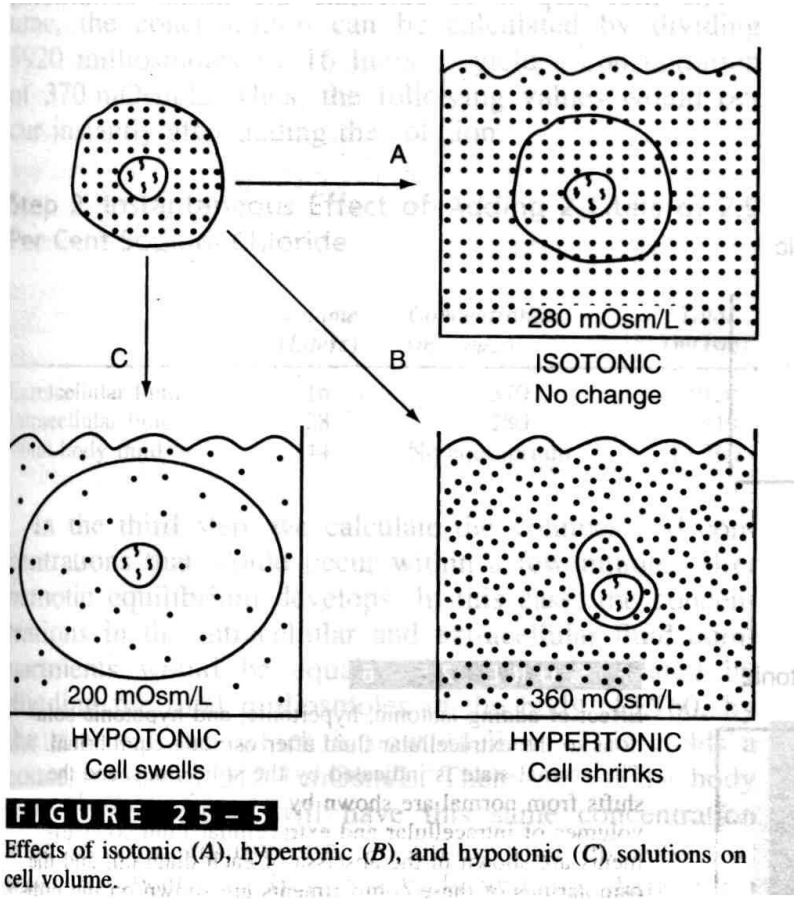


FIGURE 25–5
Effects of isotonic (*A*), hypertonic (*B*), and hypotonic (*C*) solutions on cell volume.

- An isotonic solution does not causes the change in cell size.
- Water will move from a hypotonic solution to a hypertonic solution.
- Hypotonic solution causes a cell to swell.
- Hypertonic solution causes a cell to shrink.

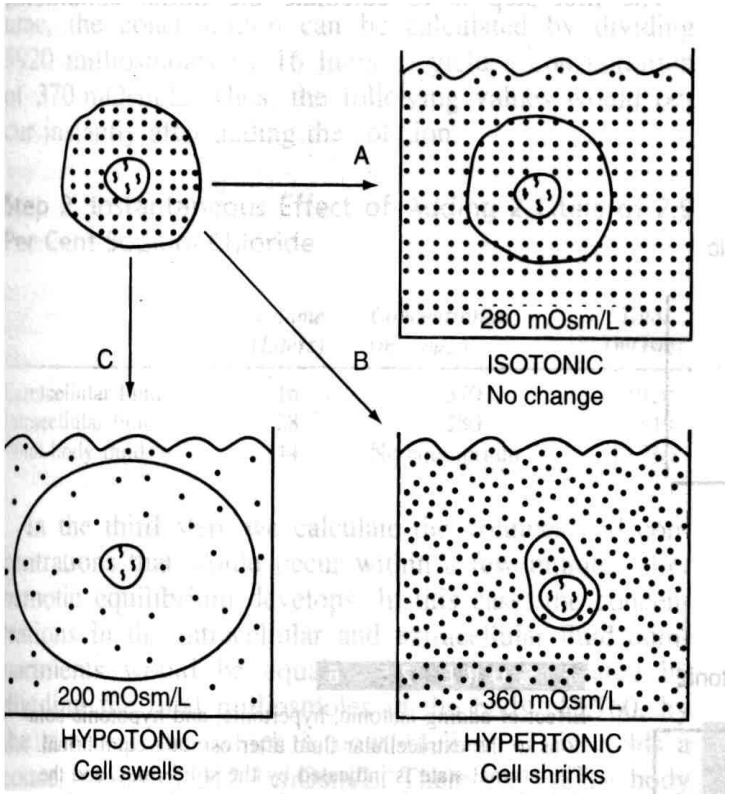


FIGURE 25-5
Effects of isotonic (*A*), hypertonic (*B*), and hypotonic (*C*) solutions on cell volume.

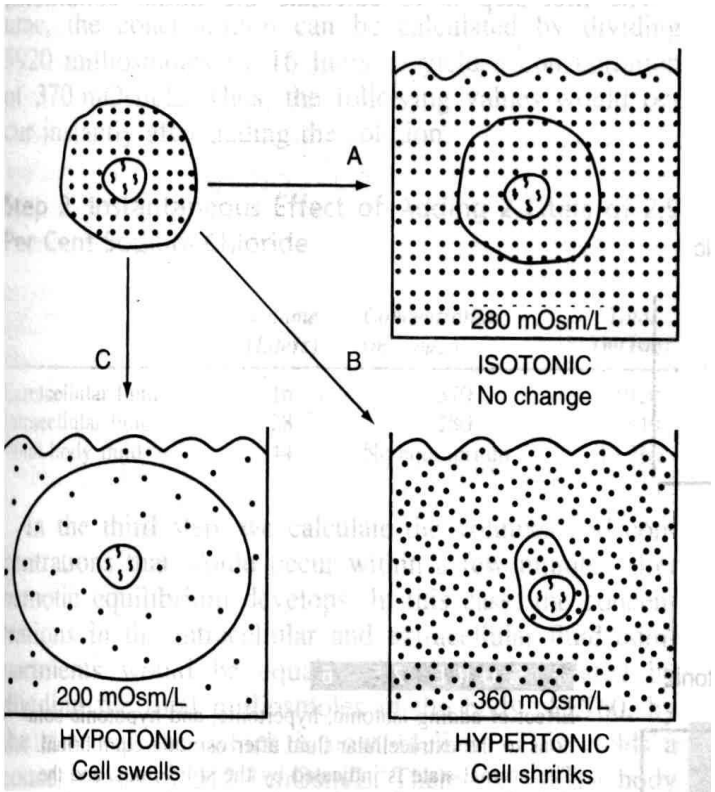


FIGURE 25–5

Effects of isotonic (*A*), hypertonic (*B*), and hypotonic (*C*) solutions on cell volume.

- The cell membrane is highly permeable to water but restricts the movement of solute molecules.
- The movement of electrolytes largely depends on the activity of transport proteins (channel, Na^+-K^+pump) in the (fat-soluble) cell membrane.

H_2O exchange across capillary

- crystal osmotic pressure??
- colloidal osmotic pressure??

2.Function of water

- (1) Acts as a medium and reactant for chemical reactions.
- (2) Acts as a transport vehicle for nutrients, electrolytes, blood gases and metabolic wastes.
- (3) Importance for the regulation of body temperature.
- (4) The water can absorb more heat with less increase of body temperature. (specific heat)
- (5) More heat can be eliminated with sweat. （enthalpy of vaporization）
- (6)Heat within the body is easy to be eliminated from the surface because of the high mobility of water.

- (7)Acts as lubricant (joint fluid in joint cavity)
- (8) Acts as a cushion (CSF, amniotic fluid for fetus)

- A man with hunger strike can survive for 1 months.
- A man with “water strike” can only survive for about 10 days.

3. Gain and loss of water

Water intake and water loss must be balanced in order to keep the volume of the body fluid constant and to maintain the homeostasis.

gains (ml/day)		loss (ml/day)	
drink	1200	lung	300
food	1000	skin	500
metabolic		feces	200
water	300	urine	1500
total	2500		2500

(1) Gain of water

- The major source of body fluid is the oral ingestion of water or soup in food.
- Another source is the water formed within the body from the metabolism of carbohydrate, protein and fat (catabolism), which is called metabolic water (endogenous water).

gains (ml/day)		loss (ml/day)	
drink	1200	lung	300
food	1000	skin	500
metabolic		feces	200
water	300	urine	1500
total	2500		2500

Catabolism

is unavoidable.

- 1g of protein generates 0.41ml of water.
- 1g of glucose generates 0.60 ml of water.
- 1g of fat generates 1.07 ml of water.

gains (ml/day)		loss (ml/day)	
drink	1200	lung	300
food	1000	skin	500
metabolic		feces	200
water	300	urine	1500
total	2500		2500

(2) Loss of water

- Water is lost from the body in several pathways.

1) Urine is the major pathway to loss the water. The volume of urine per day is at least 500 ml for eliminating waste substances of metabolism.

- There will be 35 g of urea produced each day. The urea should be eliminated at most in 6~8 % of concentration in urine, so the 500 ml of urine is needed at least each day. More urea in the body is poisonous.

gains (ml/day)		loss (ml/day)	
drink	1200	lung	300
food	1000	skin	500
metabolic		feces	200
water	300	urine	1500
total	2500		2500

2） The expired air contains water vapour.

The 300 ml of loss from lung is unavoidable in normal respiration.

Pure water is expired by lung.

gains (ml/day)		loss (ml/day)	
drink	1200	lung	300
food	1000	skin	500
metabolic		feces	200
water	300	urine	1500
total	2500		2500

- 3) The insensible vapor from skin loses 500ml of pure water per day. It is unavoidable.
- Sweat is not insensible evaporation.
- Sweat is avoidable.
- Sweat is not pure water. Sweat contains 0.2~0.3% of NaCl.(salty)
- 4) A small amount (100~200 ml) of water is normally lost in the feces.
-

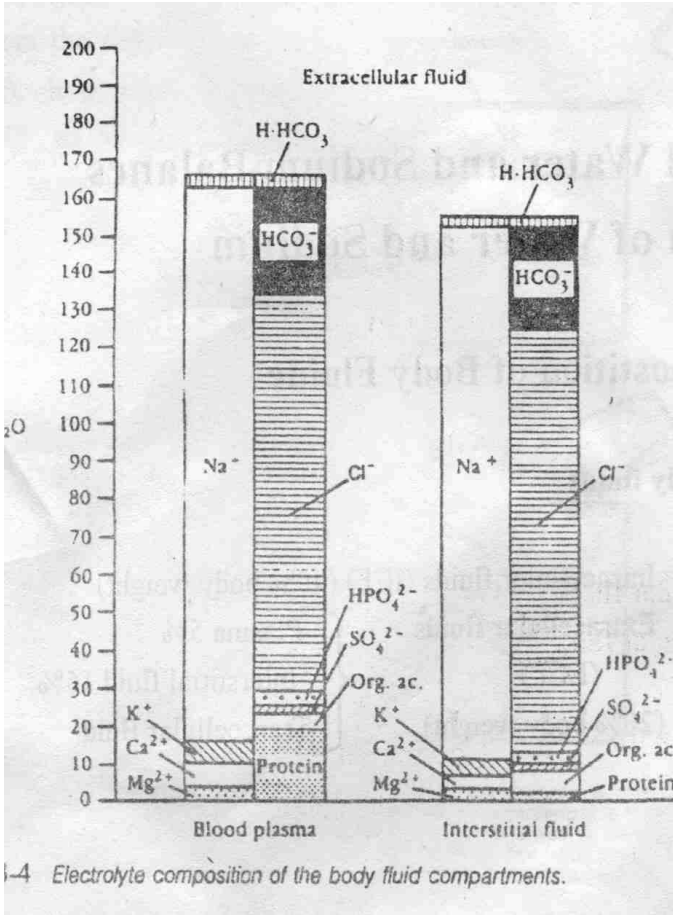
gains (ml/day)		loss (ml/day)	
drink	1200	lung	300
food	1000	skin	500
metabolic		feces	200
water	300	urine	1500
total	2500		2500

- Generally speaking, for an adult, the loss of water from skin, lung, gastrointestinal tract and kidney is at least 1500 ml per day, so 1500 ? ml of water should be replaced into the body via different ways

gains (ml/day)		loss (ml/day)	
drink	1200	lung	300
food	1000	skin	500
metabolic		feces	200
water	300	urine	1500
total	2500		2500

4. Function of sodium

- (1) Maintain the osmolality of ECF. Sodium and its attendant anions (Cl^- and $HCO3^-$) account for 90% to 95% of osmotic pressure in ECF.
- The effect of Na^+ cannot be replaced by K^+, Ca^{2+}or Mg^{2+}.



Electrolyte composition of the body fluid compartments.

- (2) Determine the membrane potential and neuromuscular excitability.

- (3) Influence the acid-base balance. $NaHCO_3$ is alkalinity, NaCI is acidity after infusion into the body. (neutrality outside the body)

- (4) Participate the metabolic process. Sodium influences the activity of enzymes.

5. Balance of sodium

(1) Content of sodium

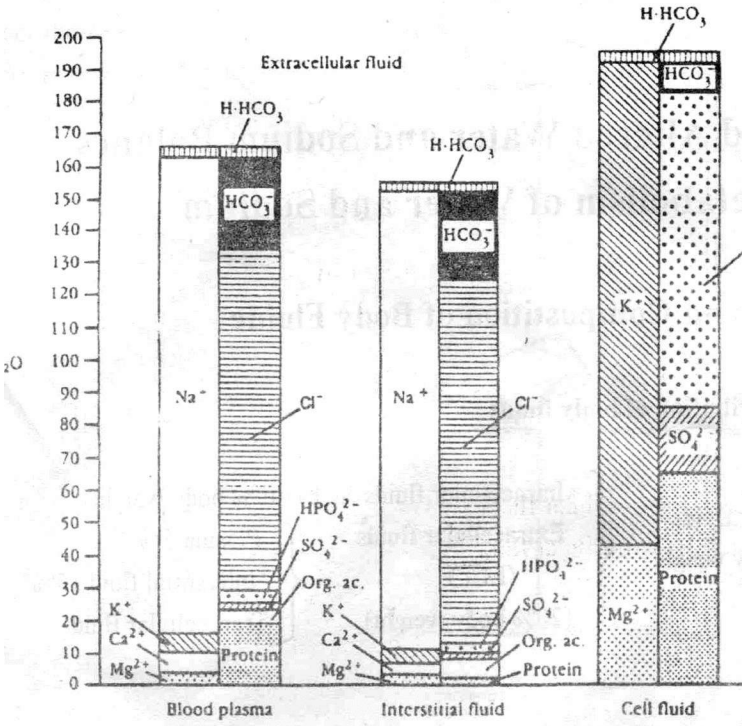
- The content of sodium is 40~50 mmol /Kg of BW.
- The sodium can be divided into two parts:
- 1)Exchangeable pool

Exchangeable pool includes sodium in extracellular fluid (50%) and intracellular fluid(10%). 60% of sodium is exchangeable.

2)non-exchangeable pool.

40% of Na^+ is bound within bone mass which is non-exchangeable.

- The concentration of Na^+ in ICF, ($[Na^+]i$) is 10 mmol/L.
- The concentration of Na^+ in ECF, ($[Na^+]e$) is 130~150 mmol/L.
-



Electrolyte composition of the body fluid compartments.

(2) Na^+ balance of intake and loss

- Body needs 500mg each day. Average daily intake from food in China is 6~15g/day. The Chinese diet contains too much sodium, especially in north China.

- Increased Na^+ in the body (H_2O and Na retention) is the precipitating factor of hypertension.

Loss of sodium

- (1)The kidneys are extremely efficient in regulating sodium balance.
- The kidneys will eliminate more if the intake of sodium is more.
- The kidneys will eliminate less if the intake of sodium is less.
- The kidneys will not eliminate sodium if there is no intake of sodium.
- (2)Gastrointestinal tract is the second way to lose sodium. Less than 10% of the sodium intake is lost from gastrointestinal tract (with feces).

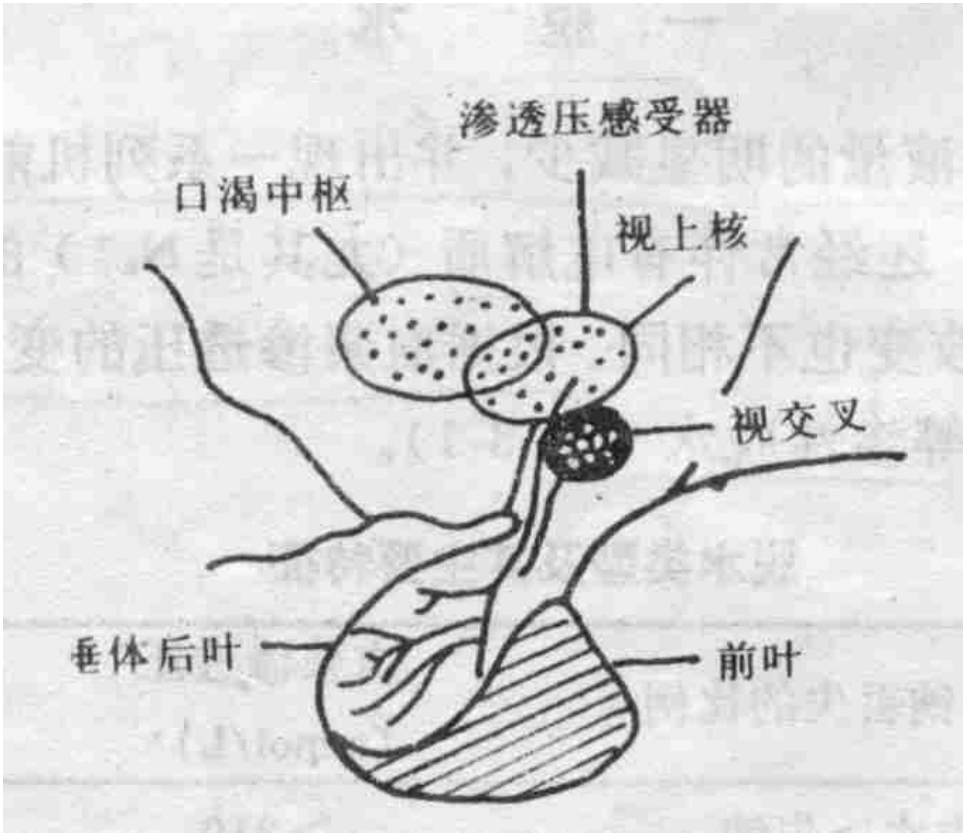
6. Regulation of water and sodium metabolism

- Normal metabolism of water and sodium means:
- (1) normal volume,
- (2) normal composition
- (3) normal osmotic pressure of body fluid.
- It is regulated by nervous system and hormones:

- (1) Thirst (neuro-regulation)
  (2) Antidiuretic hormone, ADH
- (3) Aldosterone
- (4)Atrial natriuretic peptide (ANP).
- (5) Renal regulation

(1) Thirst (neuro-regulation)

Thirst is a kind of sense (feeling) coming from thirst center, with which we will ask for drinking water.

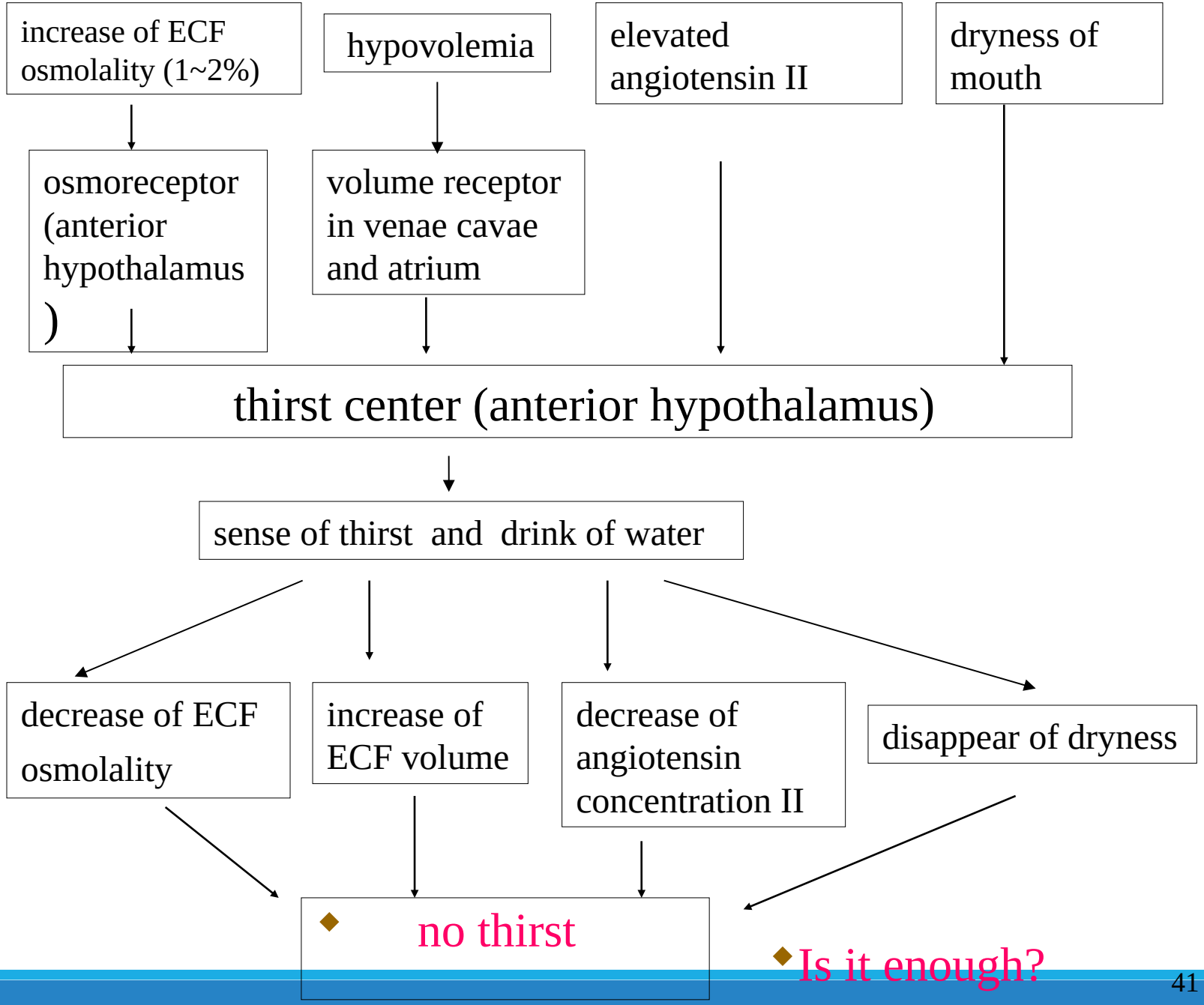


Causes of thirst:

1)Increase of ECF osmolality (1~2%) stimulates the thirst center via osmoreceptor in anterior hypothalamus. Then there will be the sense of thirst.

- After the drink of water, the increased ECF osmolality will decrease to normal, then the sense of thirst will disappear.

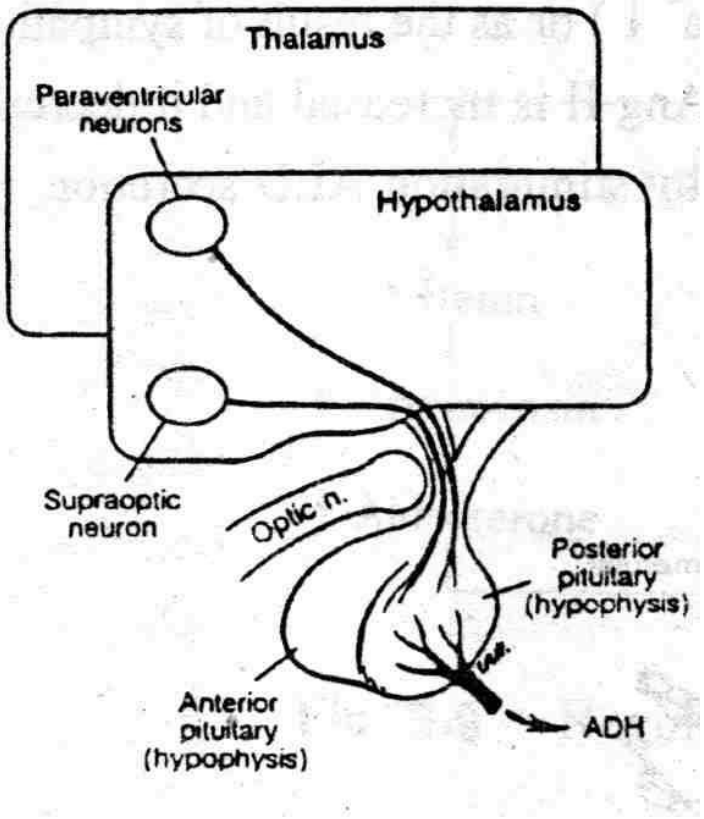
- 2) Decrease of effective blood volume (hypovolemia) stimulates the thirst center via volume receptor in venae cavae and atrium.
- 3) Elevated level of angiotensin II can stimulates the thirst center.
- (Hypovolemia→renal ischemia→renin release→angitensin I→ angiotensin II)
- 4) Dryness of mouth can also stimulates the thirst center.
- After the drinking of water, the blood volume will increase to normal.


increase of ECF osmolality (1~2%)
hypovolemia
elevated angiotensin II
dryness of mouth
osmoreceptor (anterior hypothalamus)
volume receptor in venae cavae and atrium
thirst center (anterior hypothalamus)
sense of thirst and drink of water
decrease of ECF osmolality
increase of ECF volume
decrease of angiotensin concentration II
disappear of dryness
no thirst
Is it enough?

- Thirst only is not enough to regulate the balance of water and sodium.
- The defects of (neuro-regulation) thirst are:
- 1)There will be no obvious thirst in patients with hypovolemia if the osmolality is not increased.
- 2)A patient with coma will not have the sense of thirst.

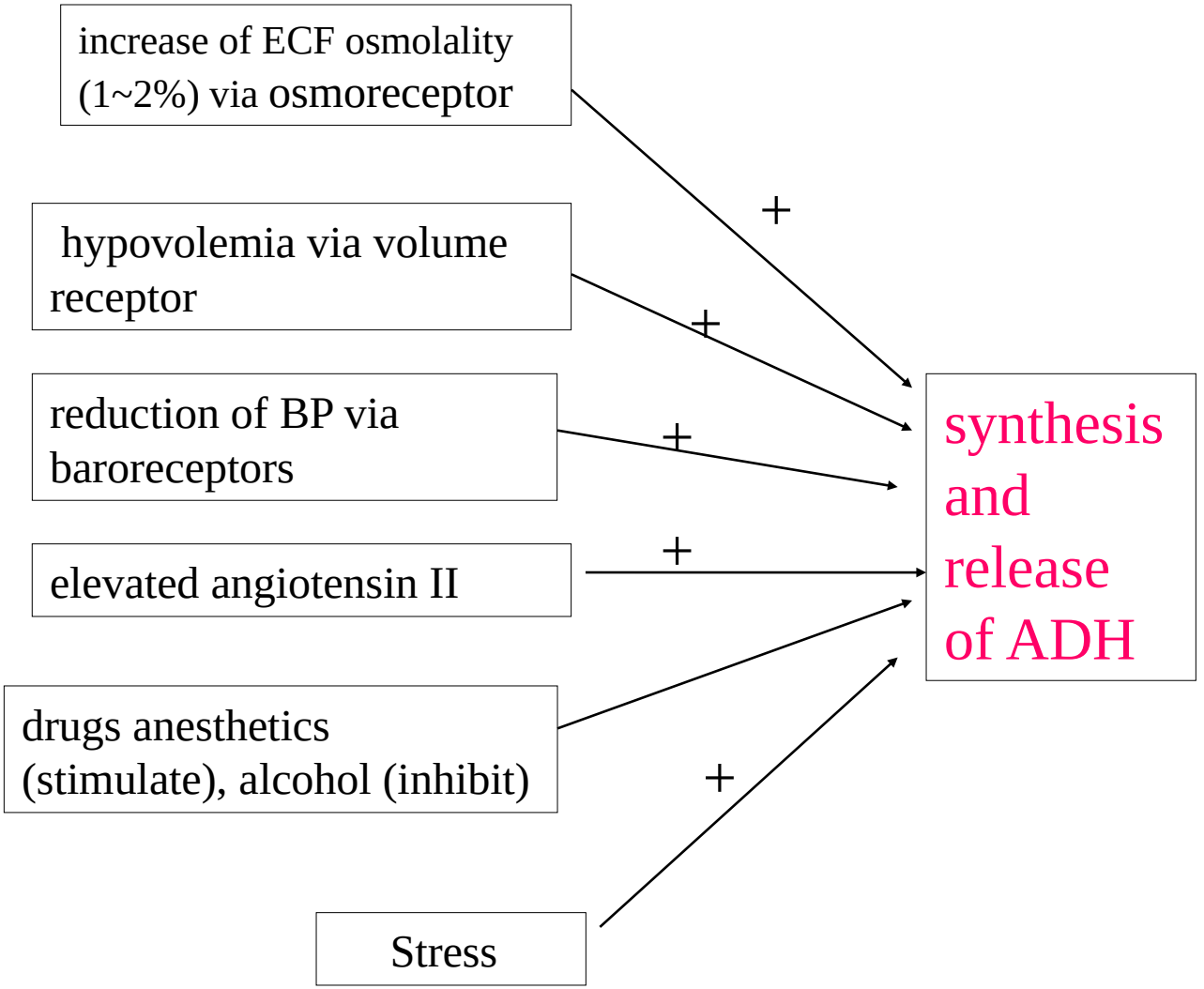
Hormoral regulation
(2) Antidiuretic hormone, ADH

- ADH is synthesized by cells in the supraoptic and paraventricular nuclei of hypothalamus, and stored in and released from posterior pituitary.
-

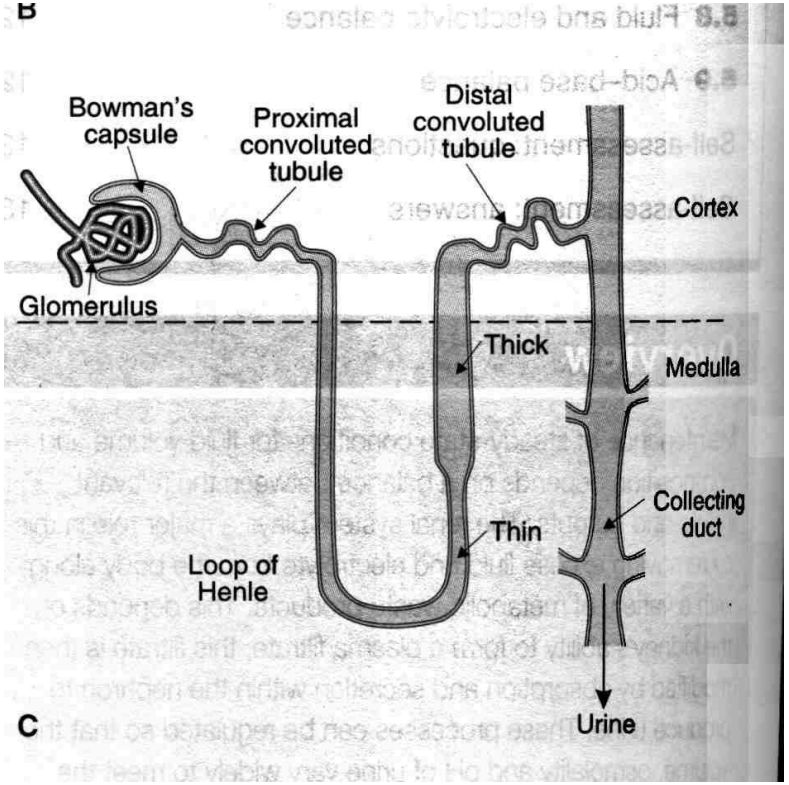


The following factors will increase the secretion of ADH.

- 1) Increased osmolality of ECF(1% ～ 2%) increases ADH secretion by stimulating the osmoreceptor (in hypothalamus).
- 2) Decreased volume of ECF （5% ～ 10%） via volume receptor (in venae cavae and atrium).
- 3) Reduction of blood pressure via baroreceptors (in arch of aorta and carotis)
- 4) Stress situations (patients during surgery, severe trauma and pain)
- 5) Increased angiotensin Ⅱ
- 6) Some drugs: anesthetics (stimulate), alcohol (inhibit)


increase of ECF osmolality (1~2%) via osmoreceptor
hypovolemia via volume receptor
reduction of BP via baroreceptors
elevated angiotensin II
drugs anesthetics (stimulate), alcohol (inhibit)
Stress
+
+
+
+
+
synthesis and release of ADH

- ADH is the major regulator for water output and reabsorption in kidney.
- A rise of ADH increases the permeability of the tubular wall to water, then increases the reabsorption of water from the distal tubules and collecting dusts of the kidneys.



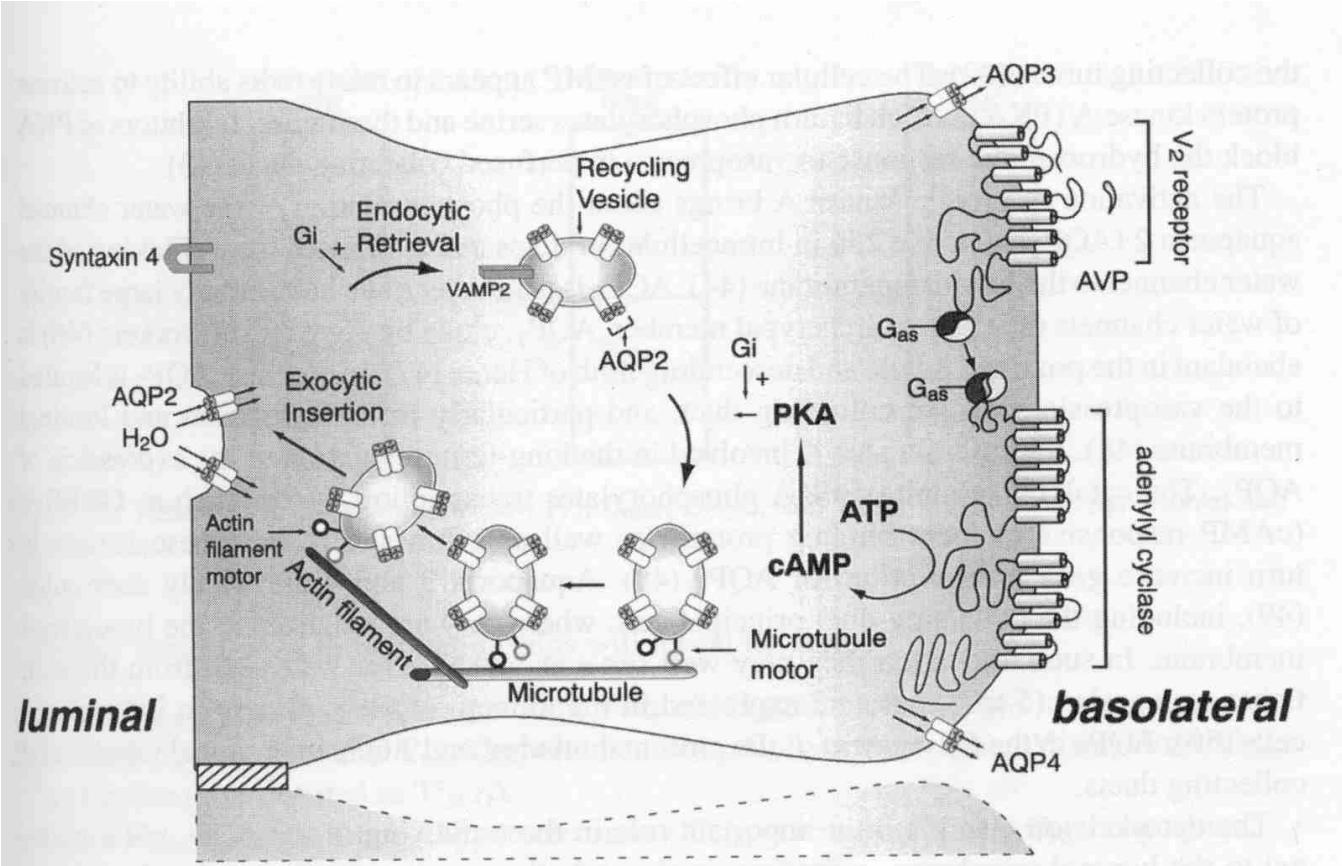
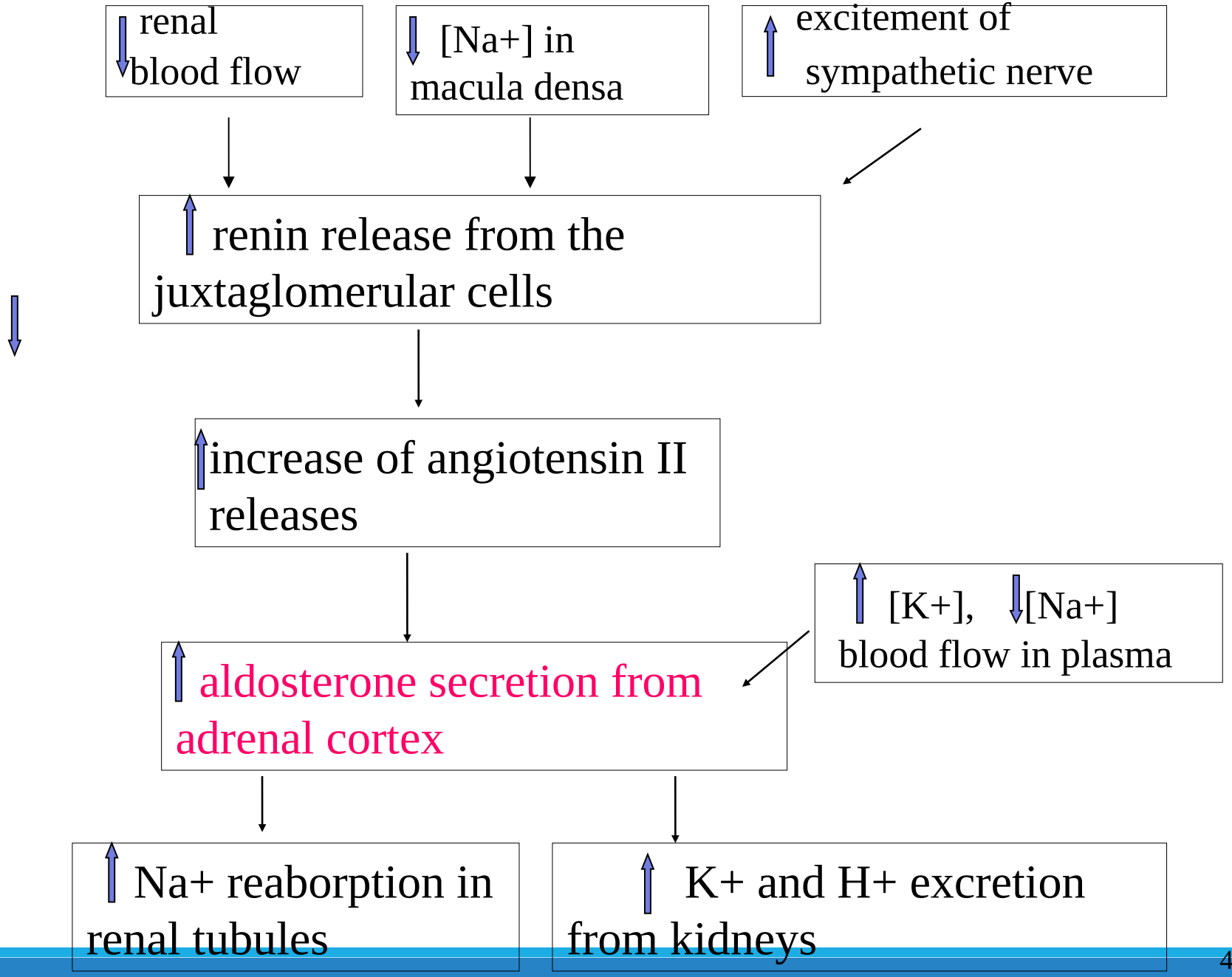
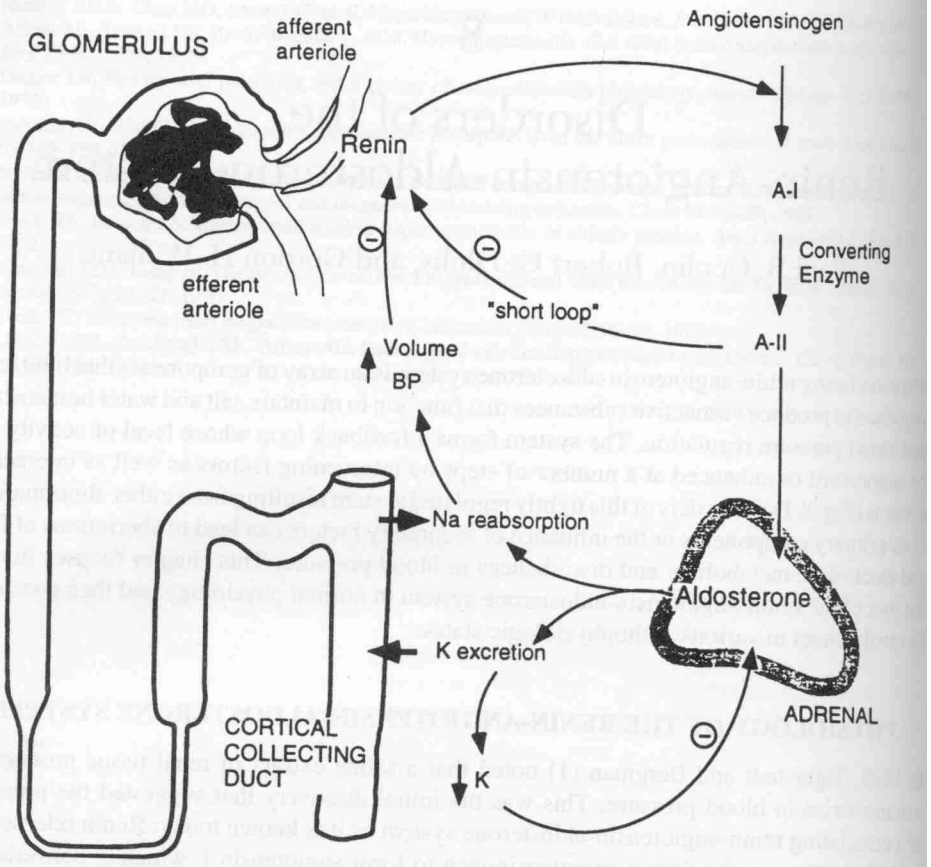


FIG. 1.10. Schematic representation of the cellular action of vasopressin. The binding of vasopressin to the V_2 receptor in the basolateral membrane initiates a cascade of events resulting in aquaporin–2 insertion in the luminal membrane, see text for details. (From Bichet D. Nephrogenic and central diabetes insipidus. In: Schrier RW, ed. *Diseases of the kidney and urinary tract,* 7th ed, vol 3. New York: Lippincott Williams & Wilkins, 2000:2553, with permission.)

(3) Aldosterone (hormoral regulation)

- Aldosterone is the major regulator of sodium excretion and reabsorption.
- Aldosterone is a steroid hormone produced by the adrenal cortex. The following factors are known to stimulate the secretion of aldosterone.
- 1) Angiotensin II is the major stimulator.
- 2) A decrease of sodium level in ECF will stimulate aldosterone secretion.
- 3) An increase of potassium level in ECF will stimulate aldosterone secretion.


renal blood flow
[Na+] in macula densa
excitement of sympathetic nerve
renin release from the juxtaglomerular cells
increase of angiotensin II releases
[K+], [Na+] blood flow in plasma
aldosterone secretion from adrenal cortex
Na+ reaborption in renal tubules
K+ and H+ excretion from kidneys



G. 8.1. The renin-angiotensin-aldosterone system is a feedback loop that functions to maintain circulat- blood volume. It can be enhanced or suppressed at various steps by other volume-regulating factors. angiotensin; BP, blood pressure. (From Mortensen RM, Williams GH. Aldosterone action. In: DeGroot et al, eds. *Endocrinology,* 3rd ed. Philadelphia: WB Saunders, 1995:1669, with permission.)

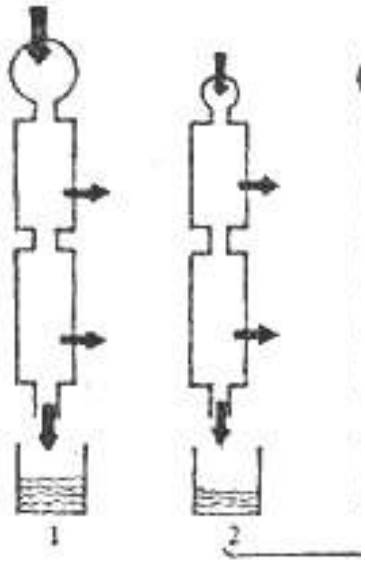
（4） Atrial natriuretic peptide(ANP)

- ANP is a 28-amino acid peptide released from the atrium in response to increased atrial stretch via mechanoreceptors.
- The effect of ANP are to:
- 1)dilate the renal blood vessels to increase the GFR,
- 2) inhibit the reabsorption of Na^+ from collecting dusts..
- 3) inhibit the release of renin, aldosterone and ADH.
- 4) be a endogenous antagonist to angiotensin II.
- The result is to increase the kidney’s ability to excrete both water and sodium.

(5) Renal regulation

- 1) Glomerular filtration
- 2) Reabsorption and excretion of renal tubules

Glomerular filtration



GFR= 125ml/min=180L/day

GFR depends on:

(a) filtration pressure

(b) permeability of glomerular membrane

(c) surface area of filtration.

60~70% of filtrate is reabsorbed at proximal tubules, finally 99%~99.5% of filtrate is reabsorbed by renal tubules, which is called glomerular-tubular balance.

Summary of regulating water and sodium metabolism:

(1) Neuro-regulation

- Thirst

(2) Hormoral regulation

- ADH
- Aldosterone
- ANP

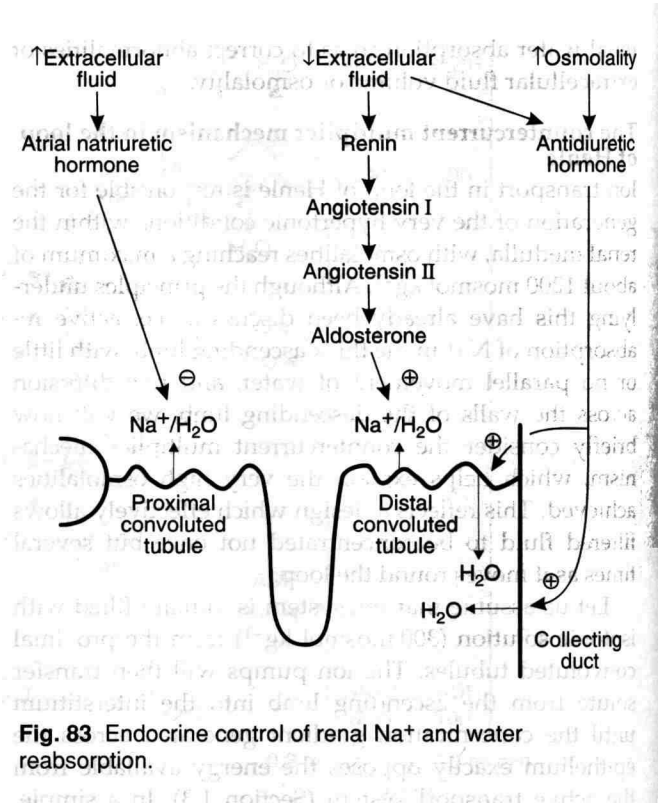


Fig. 83 Endocrine control of renal Na^+ and water reabsorption.